

Accelerating design-technology co-development using neural compact modeling and data-driven SPICE simulation.

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Abstract— This paper proposes a new design-technology co-optimization framework that expedites circuit optimization by utilizing the neural compact modeling (NCM) and a data-driven SPICE simulation. An efficient retargeting strategy of NCM and its improved design capability through a direct data driven SPICE simulation were leveraged at the industry level in response to increasingly challenging current development situations. To facilitate rapid feedback for extensive trial and error in technology optimization, the NCM swiftly fine-tune itself using pre-trained model. Then, the data interpolation and derating techniques are utilized to provide the same design environment as before such as instance binning, process variations, and layout dependent effects. Demonstrating the robustness of our framework, we achieved a 95% reduction in PDK release time while maintaining model consistency and performance at a mid-scale design of 15k transistors, with no SPICE run time and accuracy loss. This solution allows for rapid incorporation of process changes into the design, supporting quick path-finding during a design-technology co-development.

Keywords—Neural compact model, Data-driven SPICE model, Design-technology co-optimization

I. INTRODUCTION

In recent years, there has been an increasing demand for the concurrent development of technology and design, that chip design and technology optimization are carried out in tandem rather than sequentially. This approach aims to streamline development and improve product performance in the face of today's increasing complexity of technology, which necessitates extensive design explorations and Design-technology co-optimization (DTCO). Although rapid feedback of process changes to circuit designers is important for quick path-finding in this stage, the slow turnaround time (TAT) for model parameters extractions (MPE) of compact models for the SPICE simulation is a major bottleneck in such co-development efforts. A compact model is a crucial component that bridges technology and design by implementing device characteristics in SPICE simulations for integrated circuits. When device characteristics change, a new model must be developed through MPE. However, because hundreds of model parameters (MPs) in conventional model such as BSIM [1] need to be adjusted, this process is time-consuming and tedious. As a result, frequent process changes can lead to significant delays in product development schedules due to the long TAT of MPE.

So far, efforts have been made to automate the MPE of conventional equation-based compact models using genetic algorithms [2] or deep learning [3, 4] to reduce the MPE's

TAT. However, these algorithms face challenges in defining which parameters to use, determining their sequence and range. Also, managing hundreds of MPs impose complexity and limitation on achieving model accuracy. Therefore, fully automating the development of traditional equation-based model optimization remains limited due to the involvement of numerous MPs.

To solve such limitation, we applied the Neural Compact Model (NCM) as a solution of rapid development of SPICE model while reflecting variation of device characteristics effectively. Recently, the compact model community has begun to pay attention to the development of NCM for high fitting accuracy of emerging devices [5-8] and fast fitting capabilities because of MPE being unnecessary. Wang and co-workers [5] adopted the conversion function at the output layer of the NCM to make drain current $I_D = 0$ at drain voltage $V_{DS} = 0$ and effectively train the significant order of magnitude difference between the sub- and above-threshold region. Tung et. al. [6] validated I_D characteristics of NCM up to 3rd-order derivatives and Jeong et. al. [7] simulated circuits based on the NCM using SPICE porting method through Verilog-A. Also, M. Cho and co-workers [8] developed the NCM by transfer learning to enhance accuracy and reduce training time for emerging organic field-effect transistors.

Although many recent studies have been developed NCM to be more physical and practical, several factors still make it challenging to apply in the industrial domain. First, the model retargeting to the selected electrical targets (ETs) has been rarely studied to construct the new model. Normally, when device characteristics change due to process modifications in the fab, representative ETs that reflect new device properties are used to retarget a new model instead of the full I-V and C-V curves to reduce TAT of the new model development. Although Wang et. al. [6] suggested ETs targeting by adjusting the last hidden layer of NCM, it was presented without sufficient validation of the model's consistency across the entire operating regime except for specific targets. Additionally, NCM still lacks design capacity compared to the existing design framework like equation-based compact models. Further research is necessary to address not only ET retargeting but also essential aspects of IC design such as binning model across design input variables, managing process variation, and developing layout-dependent models (LDEs).

In this work, we propose an efficient NCM self-retargeting strategy by transfer learning method with new input ETs. Furthermore, we introduce a direct data-driven SPICE

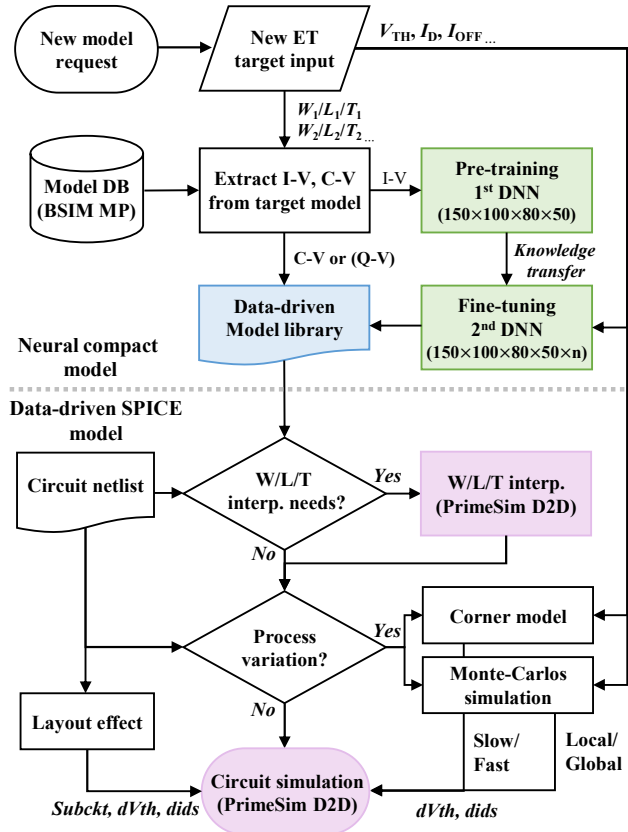


Figure 1. Workflow overview from the self-retargeting of the NCM to an enablement of the data-driven SPICE model.

simulation framework, DataSPICE, exploiting the commercial SPICE simulator PrimeSim [9]. This allows fast and accurate analysis for design-technology co-development by combining NCM and various design capabilities through data interpolation and derating techniques of the trained dataset.

II. WORKFLOW OVERVIEW

Fig. 1 illustrates the workflow of the proposed NCM and data-driven model-based design framework presented in this paper. When device characteristics fluctuate due to technological changes, such as ion implantation or modifications of device structure, and a new SPICE model is required, process engineers provide new ETs such as threshold voltage (V_{TH}), I_D and off current (I_{OFF}). In traditional equation-based compact model frameworks, retargeting the model to new ETs involves adjusting hundreds of MPs from previously published models in database (DB). This process is time-consuming and can take several weeks, depending on the model.

We developed the self-retargeting method of the NCM to reduce TAT of the new model development. First, we pre-trained the I-V and C-V (or Q-V) datasets of the reference model extract from the DB using a Deep Neural Network (DNN). We then apply knowledge transfer to fine-tune the model for the new ETs. Finally, we can create a data-driven model library based on the retargeted I-V and C-V datasets. If there are only DC variations due to technology changes irrelevant to capacitance, the dataset of C-V are directly generated from the reference model, considering only the V_{TH} shift, without additional training. In this work, we consider this situation so that only I-V datasets are used for self-retargeting of NCM.

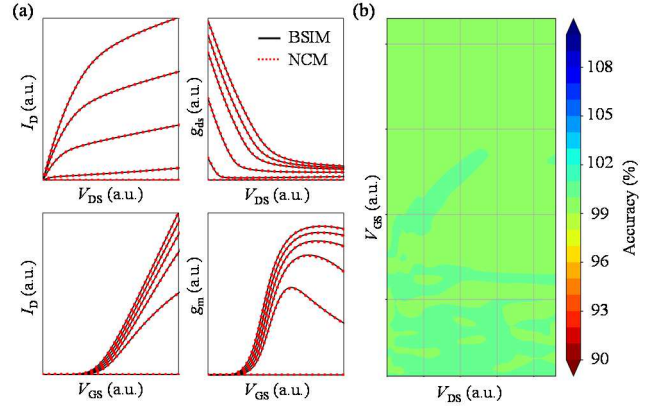


Figure 2. Pre-training result of NCM (red dotted line) versus the BSIM (black solid lines).

Once the data model library is ready, we perform direct data-driven SPICE simulation of the circuit netlist using the PrimeSim D2D [9]. To date, most NCMs have used Verilog-A to port SPICE and the drain current DNN models, but it poses difficulties in being compatible with conventional MPs-based design frameworks, such as binning, process variation and LDEs. By leveraging the fact that the inputs and outputs of the NCM are dataset, we employ data interpolation and derating techniques to integrate seamlessly into existing design environments. First, if design variables defined in the netlist such as width (W), length (L), and temperature (T) are not present in the input dataset, D2D performs interpolation internally with respect to $W/L/T$. Additionally, if the netlist requires process variation, we can adjust the dataset to slow or fast corners using data derating or generate normal distributions of V_{TH} and I_D to perform Monte-Carlo based statistical simulations. If consideration of LDEs is necessary, sub-circuit (subckt) and data derating enable to combine custom models that emulate LDEs.

III. RETARGETING STRATEGY OF NCM

The NCM consists of two training steps: The pre-training I-V characteristics of the reference model in the 1st DNN and the fine-tuning it to new ETs through the 2nd DNN using the transfer learning. We extract sequential dataset of I_D with respect to four input variables of NCM, gate (V_{GS}), drain (V_{DS}), body (V_{BS}) voltage and T . The bias range of V_{GS} and V_{DS} is selected to be sufficiently large to cover V_{DD} of target circuits, with steps of approximately 30 to 70 mV while V_{BS} is set with a wider interval of 100 to 300 mV. T is split into 3 different conditions, cold, room and hot temperature. Finally, we have a dataset consisting of over 30,000 I_D points per transistor and use it to train the 1st DNN with four hidden layer with different number of neurons (150, 100, 80, 50) and one output layer for I_D . We adopt conversion function $I_D = I_0 \cdot V_{DS} \cdot 10^V$ [6] and the mean absolute percentage error (MAPE) loss function of I_D , g_m , g_{ds} for training as follows:

$$Loss = a \cdot MAPE(y) + b \cdot MAPE(g_m) + c \cdot MAPE(g_{ds}) \quad (1)$$

The 1st DNN for the pre-training uses optimal constants in loss function, $a = 1$ and $b = c = 0.1$ to get high consistency for all I_D , g_m and g_{ds} . The model includes the adaptive moment estimation ("Adam"), tanh as the activation function and trains 150,000 epochs to capture full dataset. To evaluate pre-training consistency of NCM, we examine a dataset of n-type planar field-effect transistor. Fig. 2 shows that characteristics

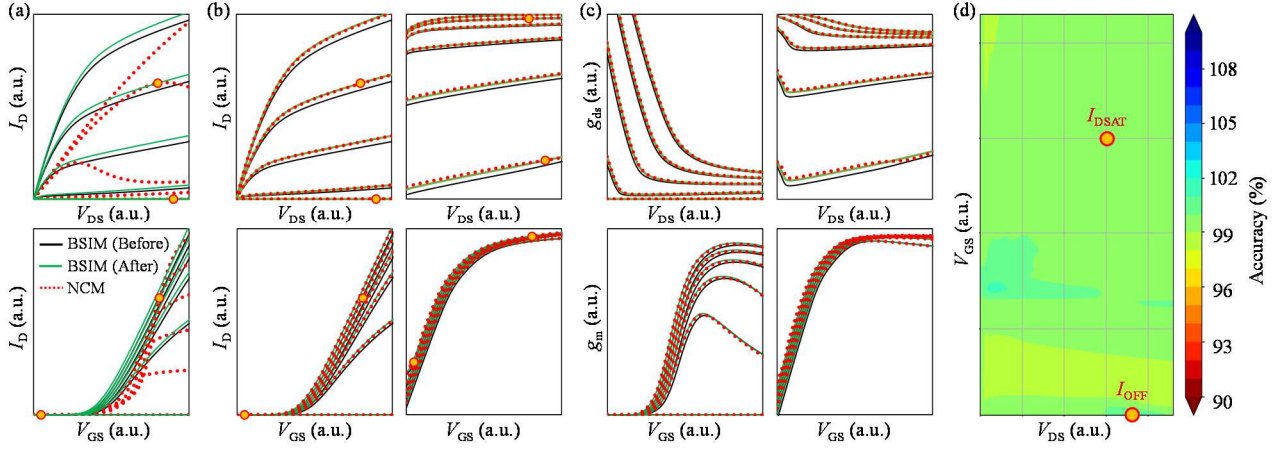


Figure 3. Fine-tuning result of NCM (red dotted lines), referenced BSIM for both before (black solid lines) and after (green solid lines) retargeting with 2 ETs case. (a) denotes result with $a = 1$, $b = c = 0$ in (1), (b-d) shows result with $a = 2$, $b = c = 1$ in (1) for I_D , g_m , g_{ds} and 2D plot of I_D regarding V_{DS} and V_{GS} .

of I_D , g_m and g_{ds} are fully captured by DNN. Particularly, 2D accuracy plot for I_D with respect to V_{GS} and V_{DS} is present, calculating the accuracy on a linear scale in the above-threshold region and on a logarithmic scale in the sub-threshold region using V_{TH} as the boundary. This validates that the consistency of the model is effective in the sub-threshold regime despite its significantly large order difference compared to the above-threshold region. In addition, the derivatives of I_D , g_m and g_{ds} are also precisely trained in the entire bias regime.

After pre-training, we construct 2nd DNN by knowledge transfer to retarget input ETs while retaining the pre-trained device properties in the 1st DNN. The 2nd DNN adds a new 5th hidden layer (150, 100, 80, 50, n) while weights and bias in first four hidden layers of 1st DNN are fixed. The number of neurons in the new layer, n , equals to number of input ETs to prevent the divergence of the model during fine-tuning. Here, we examine 2 input ETs case for I_{DSAT} ($V_{DS}=V_{GS}=V_{DD}$) and I_{OFF} ($V_{DS}=V_{OFF}$, $V_{GS}=0$ V), therefore $n = 2$. The 2nd DNN also adopts ADAM estimation, tanh activation function and MAPE as loss function similar to the 1st DNN. The key difference of the second DNN is that the dataset of ETs for retargeting contains normally fewer than 10 ETs, which is significantly smaller compared to the pre-training. Therefore, if the fine-tuning considers only this small ETs dataset, it is difficult to expect accuracy in regions outside the ETs due to overfitting (Fig. 3a). Although the I_D consistency at ETs is close to 100%, the curve profile is completely deviated from the reference model.

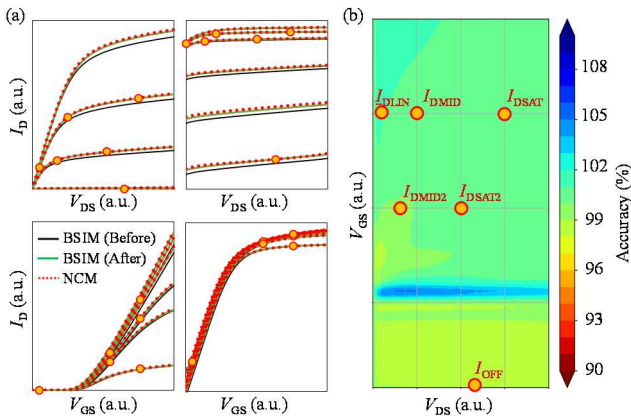


Figure 4. Fine-tuning result of NCM (red dotted lines), referenced BSIM for both before (black solid lines) and after (green solid lines) retargeting with 6 ETs case.

To solve the overfitting problem during self-retargeting, we modify the components in the loss function by considering derivatives of I_D , g_m and g_{ds} , to reflect the curve profile during the fine-tuning as well as ETs in (1). Here, the train dataset of g_m and g_{ds} are selected from the previous model before retargeting. By adjusting a , b and c in the loss function (1), this physic-informed like DNN can successfully predict a new model. We set $a = 2$ and $b = c = 1$ to give more weight on ETs than the curve profile of the previous model to hinder the model from ignoring input ETs and following the previous curve. In Fig. 3b and c, self-retargeted NCM show good accuracy in I_D and its derivatives in both linear and logarithmic scale with respect to the retargeted one based on the BSIM model. Even in 2D accuracy plot, the consistency of the self-retargeted NCM is almost perfect in all operating ranges.

In real cases, the number of ETs that need to be targeted by MPE varies depending on the model. For example, a single target is sufficient when circuits validation is enough at V_{DD} , but for circuits operating various regions, multiple ETs are necessary. In Fig. 4, we demonstrates the accuracy of self-retargeted NCM based on 6 ETs input, I_{DSAT} , I_{DSAT2} , I_{DMID} , I_{DMID2} , I_{DLIN} and I_{OFF} . The fine-tuned I-V curves properly capture all 6 ETs including linear and off regions while containing the curve profile of the reference model. The average accuracy is over 99 % at 6 targets and over 95% at other regions, which means this retargeting algorithm is reliable even for more complex input cases.

Regarding TAT of model development, self-retargeting NCM strategy has advantages over the conventional equation-based framework. In our system with new model development is nearly 100 hours for a model consisting of 10 W/L sizes configured with 6 ETs. In contrast, since the pre-training and fine-tuning of a transistor takes almost 0.5 hour based our computational condition in our NCM method, it can dramatically reduce TAT by 95%. In other words, we can achieve 20 times more activities of the design exploration and DTCO within same development period. Moreover, we can further reduce TAT if the computational capabilities improve.

IV. DATA-DRIVEN SPICE SIMULATION

A. Instance interpolation

To enhance the design capacity of NCM, we adopt a direct data-driven SPICE simulation, DataSPICE, via the commercial SPICE simulator, PrimeSim D2D [9]. It accepts

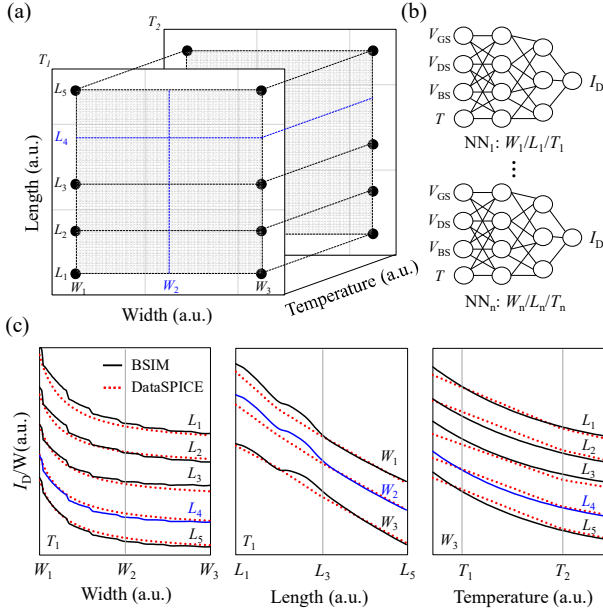


Figure 5. (a) $W/L/T$ Interpolation schematic, (b) neural-network of NCM and (c) interpolation results of conventional SPICE with BSIM (solid lines) versus DataSPICE with NCM (dotted lines) for various design variables. Blue lines denotes result of variables not included in the input dataset.

I-V, C-V (or Q-V) dataset and enables SPICE simulation by designed interpolation and extrapolation schemes. This secures the simulation speed and robustness across all bias ranges. Also, D2D provides $W/L/T$ binning of discontinuous dataset input by conducting scaling across design variables, like $W/L/T$, during dataset read-in stage of PrimeSim. In the conventional design methodology, few $W/L/T$ sizes are enough to cover a wide range of the model coverage by seamless binning of key model parameters, such as V_{TH0} and U_0 in BSIM [1]. In contrast, NCM should employ a large global model to guarantee a broad coverage. However, this approach not only incurs significant computational cost but also has limitations in accurately predicting characteristics over a wide range. Thus, we use an interpolation method to connect data from each small local model, ensuring comprehensive coverage and reducing both overhead and data size.

Fig. 5a is a schematic of the $W/L/T$ interpolation of DataSPICE. First, when we receive a new model request, NCM-based self-retargeting method generates new datasets using the ETs for the primarily used sizes in the design, marked as black dots on the $W/L/T$ plane. This process involves using one neural network (NN) for each unique $W/L/T$, such as $NN_1: W_1/L_1/T_1 \dots NN_n: W_n/L_n/T_n$ (Fig. 5b), and linearly interpolating these nodes over respective instance to implement model coverage (gray area in Fig. 5a).

Fig. 5c demonstrates the consistency of the interpolation. For various length inputs, L_1 to L_5 , the interpolation over the width coverage range from W_1 to W_3 (red dotted lines) exhibits accurate predictions compared to conventional seamless binning-based BSIM models (black solid lines). The length interpolation for width inputs, W_1 to W_3 , and the temperature interpolation at specific W/L conditions both shows high accuracy, within 3% of error rates with the reference. Notably, even for conditions such as W_2 and L_4 , which were originally not selected as input sizes but instead interpolated (blue solid lines), the interpolation results accurately matches the

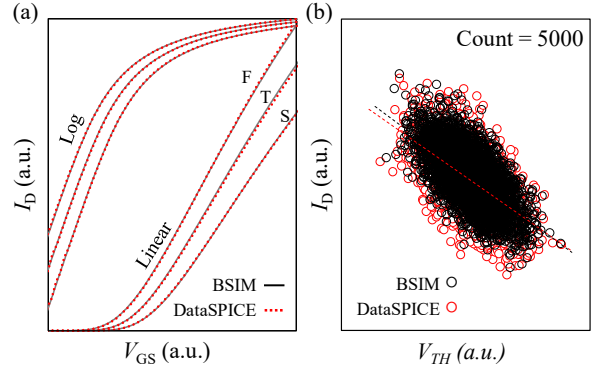


Figure 6. Two process variation result by conventional method (black) versus DataSPICE (red). (a) I_D - V_{GS} curve for slow and fast corner model in both linear and logarithmic scale and (b) V_{TH} - I_D correlation of 5000 times MC simulation.

reference data. This demonstrates that data interpolation can comprehensively ensure model coverage.

B. Process variation

D2D also supports two representative process variation models, process corner model and Monte-Carlos (MC) based statistical simulation. In fact, the data-driven model had a drawback of not being compatible with process variations, which has limited its application. To overcome this limitation, we adopt data derating techniques, provides dv_{th} and $dids$ parameters that can directly adjust V_{TH} and I_D within given dataset. In other words, these parameters can make process corners or random distribution of the input dataset.

Fig. 6a illustrates linear and logarithmic scales of the generated slow and fast corners using the derating method of a typical dataset. When requesting a new model, slow and fast corner skew for V_{TH} and I_D is normally provided. We first horizontally shift the dataset's table using dv_{th} to account for the V_{TH} skew, and then use $dids$ to achieve the final targeting for any I_D skew. Because this method tunes V_{TH} and I_D simultaneously, the result demonstrates consistent simulation performance in both sub-threshold and above-threshold region which are comparable with the original BSIM-based corner model. Moreover, from the standpoint of data size of the data-driven model, it helps model lightweight by reducing dataset size, instead of offering all corners as datasets.

The full statistical simulation through the MC model is also possible using the same dv_{th} and $dids$ derating concept by the generation of normally distributed V_{TH} and I_D from the given dataset. In Fig. 6b, 5000 times of MC simulation of input typical dataset verifies the statistical modeling in DataSPICE has similar variation coverage with the conventional model. The V_{TH} versus I_D correlation between the data-driven MC and the original BSIM-based MC simulation shows almost identical distribution, and the trend lines of both data correlations nearly coincide. The R^2 value of the V_{TH} - V_{TH} correlation between the two MC models is 0.99, while the R^2 value of the I_D - I_D correlation stands at 0.91. Through data derating method in DataSPICE, NCM completely overcomes its incompatibility with the process variation simulation.

C. Layout dependent effect

Here, we investigated the application of the LDEs on the NCM. LDEs indicate variations in electrical performance of devices depending on the surrounding layout structure. The SPICE simulation receives structural information from input netlists, which modifies the properties of device parameters

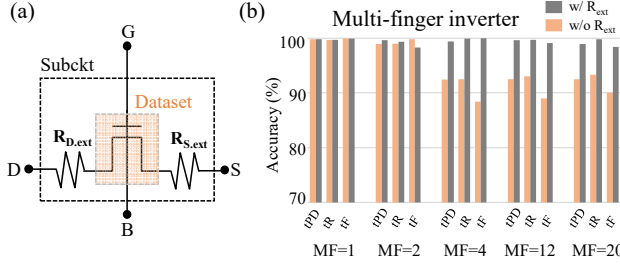


Figure 7. (a) Schematic of subckt with dataset and external resistance R_{ext} . (b) Accuracy comparison of tPD, tR, tF for various MF inverters with R_{ext} (gray) and without R_{ext} (orange).

such as V_{TH} , I_D and contact resistance (R_C). Among various LDEs, we can consider effects like gate-to-contact distance, trench isolation space and length of diffusion, which are described through variations in V_{TH} and I_D [10], into the model using the previously explained dv_{th} , did_s derating method. However, the effect that alters R_C is still challenging to integrate.

A notable example is the multi-finger (MF) devices induced LDEs to the R_C . Generally, MF devices are employed to achieve high current capabilities and layout efficiency, but their sharing junctions can lead to unexpected variations in R_C . While conventional equation-based compact models like BSIM offer internal parameters for adjusting resistance and can also be implemented using subckt, NCM has been limited because a relevant design methodology has not yet been established.

To solve this limitation, we adopt the same subckt structure of the conventional model library. Our proposed DataSPICE with PrimeSim D2D supports the subckt structure while connecting I-V and C-V (or Q-V) datasets to each device node, allowing integration with custom models used in the subckt. As shown in Fig. 7a, we attached an R_{ext} within the subckt of dataset and integrated a custom model that describes R_C variations for MF devices. By increasing MF from 1 to 20 and examining the inverter's propagation delay time (tPD), rising time (tR), and falling time (tF), we found that the error exceeded 10% without R_{ext} as MF increased. However, we reduced the error to below 1% by implementing subckt and R_{ext} as an adjustable external resistance depending on the number of MF.

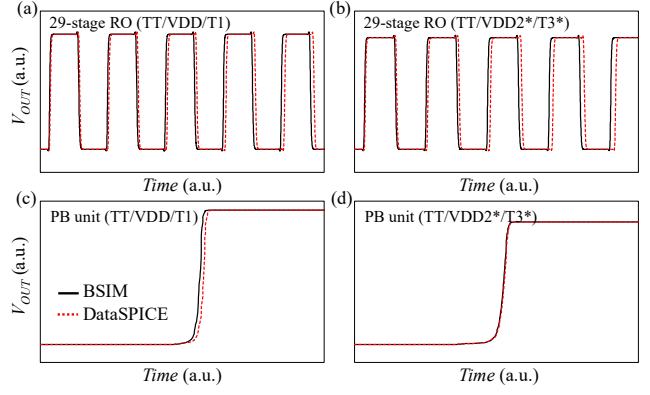


Figure 8. Circuit waveform of small-scale 29-stage RO with (a) TT/ V_{DD}/T_1 and (b) TT/ V_{DD2}/T_3 and mid-scale PB unit with (c) TT/ V_{DD}/T_1 and (d) TT/ V_{DD2}/T_3 . Black solid lines denote conventional SPICE with BSIM and red dotted lines shows DataSPICE with NCM.

V. CIRCUIT VALIDATION

We compared simulation accuracy and runtime degradation of our NCM-based DataSPICE framework against traditional BSIM-based SPICE modeling using both small-scale and mid-scale circuits. Since the data-driven model is known to encounter challenges with SPICE runtime and convergence, particularly at large-scale circuits, we evaluated circuits of two different scales to assess comprehensively. For the small-scale circuits, we used 29-stage of ring oscillator (RO) to compare with tPD, while for the mid-scale circuits, we utilized the page buffer (PB) unit of the NAND Flash Memory to compare with trip time of specific latch operation. Here, transistors used in each circuit were approximately 200 and 15000, respectively. We split the simulation conditions into 3 different process/voltage/temperature ($P/V/T$) corners to appropriately validate both interpolation and process variation. TT and SS in Table I, each represent typical and slow process corners for both n-type and p-type field-effect transistors and SS corner is generated by data derating method. Also, V_{DD} , T_1 , and T_2 are all conditions used as data inputs, whereas V_{DD2} and T_3 are not, necessitating additional interpolation with respect to V and T , respectively, to perform SPICE simulations.

Table. I presents the relative performance metrics of SPICE runtime and accuracy of the NCM-based DataSPICE with respect to the BSIM-based conventional SPICE simulation for each $P/V/T$ corner. Firstly, small scale of the 29-stage RO showed a runtime reduction of 95.6% and 98.8% as consistency of tPD exhibited over 99% under the two conditions where interpolation was unnecessary, TT/ V_{DD}/T_1 and SS/ V_{DD}/T_2 . In the case of TT/ V_{DD2}/T_3 , however, the runtime increased slightly by 102.1% due to the interpolation process while the consistency was validated at a similar level of 98.8%. For the larger-scale PB unit, the runtime also decreased to about 94% in the first two conditions without interpolation, and the consistency of trip time remained over 99%. Similar to the RO, the TT/ V_{DD2}/T_3 condition requiring interpolation showed an increase in runtime to 103.7%, with a slight decrease in consistency to 98.6%. From this result, we demonstrated that the data-driven model can perform circuit simulations without convergence issues even for mid-level scales with over 15k transistors. Notably, when interpolation was not used, the runtime decreased with remaining the original level of circuit's consistency. Demonstrating the effectiveness of our design framework, this methodology achieves notable accuracy and simulation runtime as well as

TABLE I. RELATIVE CIRCUIT RUNTIME AND ACCURACY EVALUATION OF CONVENTIONAL SPICE WITH BSIM VERSUS DATASPACE WITH NCM

Circuits	P/V/T corner	Relative runtime	Accuracy index	Relative accuracy
29-stage RO	TT/ V_{DD}/T_1	95.6 %	tPD	99.4 %
	SS*/ V_{DD}/T_2	98.8 %		99.0 %
	TT/ V_{DD2}^{**}/T_3^{**}	102.1 %		98.8 %
PB unit	TT/ V_{DD}/T_1	94.9 %	Trip time	99.3 %
	SS*/ V_{DD}/T_2	94.2 %		99.2 %
	TT/ V_{DD2}^{**}/T_3^{**}	103.7 %		98.6 %

* Data derating for process variation

** Data interpolation for not input V/T variables

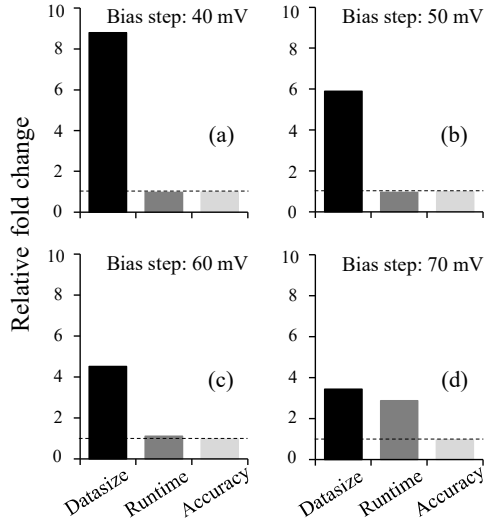


Figure 9. Relative fold change of datasize, circuit runtime and accuracy of NCM-based DataSPICE compared to conventional SPICE simulation for V_{DS} and V_{GS} step is (a) 40 mV, (b) 50 mV, (c) 60 mV and (d) 70 mV .

the robust circuit waveform (Fig. 8) comparable to the conventional SPICE simulation.

Since runtime and accuracy are inherently size-dependent in this model, selecting optimal bias steps of NCM is important. A finer dataset step reduces the number of interpolations, improving runtime and accuracy at the cost of increased data size, whereas a sparser step reduces data size but increases interpolation, causing both runtime and accuracy to deteriorate. To prove this, we constructed the model by dividing V_{DS} and V_{GS} steps in the dataset, categorized as 40, 50, 60, and 70 mV, and evaluated the PB unit with $TT/V_{DD}/T_1$ conditions. Fig. 9 shows how the data driven model's data size, circuit runtime, and accuracy changed by several folds compared to the BSIM-based conventional simulation. The data model with the finest bias step of 40mV had a data-size increase of 8.8 times, but it gradually decreased as the bias step increased, reaching 3.3 times at 70mV. The runtime was similar to the conventional model up to 60 mV, but it increased by a factor of 2.8 for 70 mV because the wider bias interval results in an increase of interpolation. Also, we confirmed that the accuracy of the simulation is consistent with conventional method in all conditions regardless of data size or runtime. Therefore, it is crucial to carefully consider the trade-off between data size and runtime when determining the step size of NCM.

VI. CONCLUSION

In this work, we proposed a NCM that enables efficiently retargets to new input ETs and its data-driven design infrastructure to guarantee equivalent design environment with the existing framework. In particular, the retargeting of device model using the transfer learning of NCM showed high consistency of I_D prediction over 95% throughout the entire operating range while constructing model 20 times faster than the conventional equation-based model. This will significantly reduce the communication delay between technology and design, allowing for more design exploration and DTCO activities within a limited timeframe and contributing to product completeness. In addition, we introduced a direct data-driven SPICE simulation, DataSPICE, to provide I-V and C-V modeling of NCM, as well as its $W/L/T$ interpolation,

process variation, and LDEs modeling. This expands the design capability of NCM and facilitates its application in practical IC chip design. We validated this framework at the mid-level circuit scale, demonstrating its utility, and enabling a rapid evaluation of DTCO's impact at a larger scale beyond the cell-level analysis. Additionally, this approach supports not only NCM but also TCAD and measurement-based datasets, facilitating swift and flexible collaboration between technology and design.

REFERENCES

- [1] J. P. Duarte, S. Khandelwal, A. Medury, C. Hu, P. Kushwaha, H. Agarwal, A. Dasgupta, and Y. S. Chauhan, "BSIM-CMG: Standard FinFET compact model for advanced circuit design," in Proc. 41st Eur. Solid-State Circuits Conf. (ESSCIRC), Sep. 2015, pp. 196–201
- [2] Yiming LI and Y.-Y. Cho, Intelligent BSIM4 Model Parameter Extraction for Sub-100nm MOSFET Era, Jpn. J. Appl. Phys., 43, 1717 (2004)
- [3] F. Chavez, C.-T. Tung, M.-Y. Kao, C. Hu, J.-H. Chen and S. Khandelwal, Deep learning-based I-V Global Parameter Extraction for BSIM-CMG, Solid State Electronics, 209, 108766 (2023)
- [4] Z. Peng, Y. Wang, Z. Yuan and X. Wang, iPREFER: An Intelligent Parameter Extractor based on Features for BSIM-CMG Models, arXiv <https://arxiv.org/abs/2404.07827> (2024)
- [5] J. Wang, Y.-H. Kim, J. Ryu, C. Jeong, W. Choi and D. Kim, Artificial Neural Network-Based Compact Modeling Methodology for Advanced Transistors, IEEE Trans. Electron Devices, 68, 3 (2021)
- [6] C.-T. Tung, M.-Y. Kao and C. Hu, Neural Network-Based I-V and C-V Modeling With High Accuracy and Potential Model Speed, IEEE Trans. Electron Devices, 69, 11 (2022)
- [7] H. Jeong, S. Woo, J. Choi, H. Cho, Y. Kim, J.-T. Kong and S. Kim, Fast and Expandable ANN-Based Compact Model and Parameter Extraction for Emerging Transistors, IEEE J. Electron Devices Soc., 11, 153 (2023)
- [8] M. Cho, M. Franot, O.-J. Lee and S. Jung, A neural compact model based on transfer learning for organic FETs with Gaussian disorder, J. Mater. Chem. C, 12, 16691 (2024)
- [9] PrimeSim User Guide: Pro and SPICE, Synopsys, CA, USA (2023)
- [10] S. Kim, S. Yang, H. Lim, H. Lee, J. Jeon, J. Choi and J. Kim, Accurate Layout-Dependent Effect Model in 10 nm-Class DRAM Process Using Area-Efficient Array Test Circuits, IEEE Access, 11, 70691 (2023)